

TorchJD: Jacobian Descent in PyTorch

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github.com/SimplexLab/TorchJD



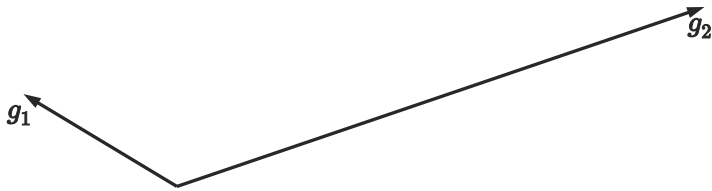
torchjd.org



arxiv.org/pdf/2406.16232

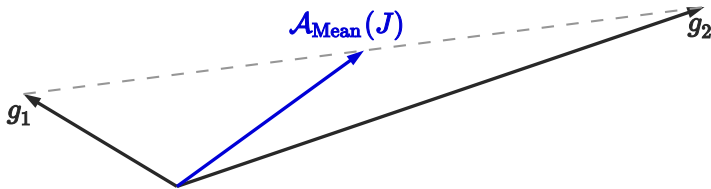
Jacobian descent

$$J = \begin{bmatrix} g_1^\top \\ g_2^\top \end{bmatrix}$$

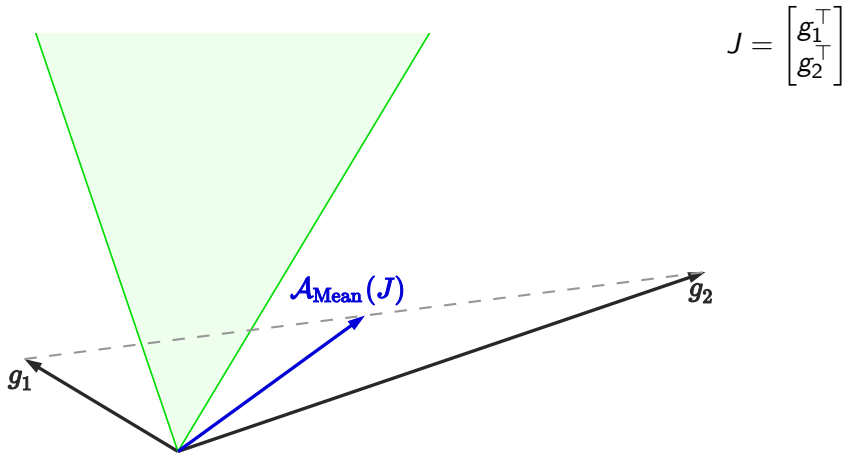


Jacobian descent

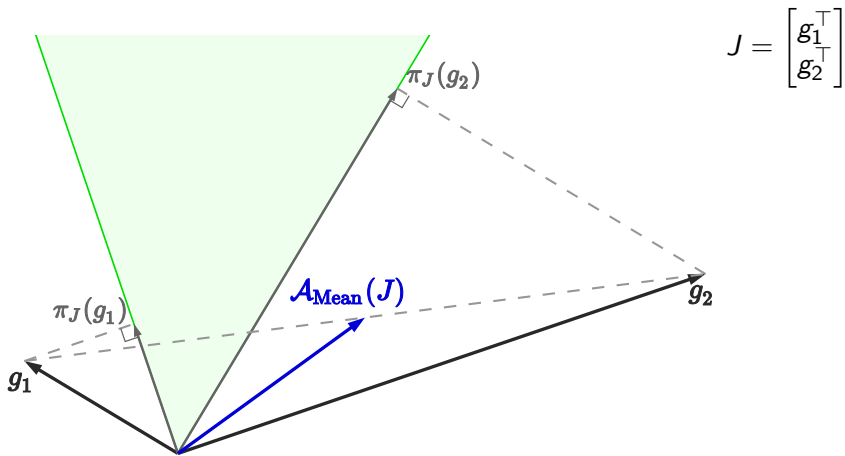
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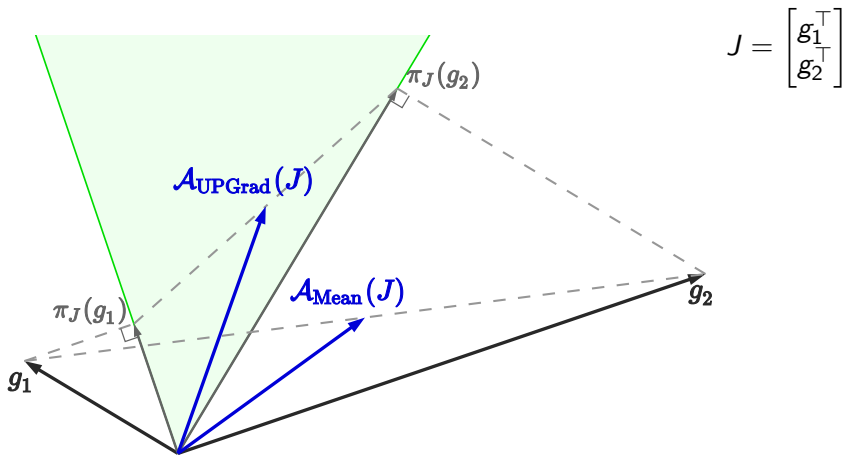
Jacobian descent



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TorchJD to compute and aggregate Jacobians.

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Try to compute the inner products between your gradients.

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Can JD solve problems that GD can't?